

Ritabrata Chakraborty

[GitHub](#) | [Google Scholar](#) | [ResearchGate](#) | [Scopus](#) | [Portfolio](#) | [LinkedIn](#) | ritabratbits@gmail.com | +91 89107 83548

EDUCATION

University of Pennsylvania

M.S.E. in Robotics, GPA: -/4

Pennsylvania, USA

Aug '26 – May '28

Birla Institute of Technology and Science, Pilani (BITS Pilani)

B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10

Rajasthan, India

Oct '22 – May '26

HONOURS & SCHOLARSHIPS

Institute Merit-cum-Need Scholarship — Consecutive Recipient, BITS Pilani

Oct '22 – May '26

PUBLICATIONS & PEER REVIEWS

Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

Poster Presentations

- R. Chakraborty, “*High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.*” Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, Visakhapatnam, India, Sep 2024.

Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

RESEARCH EXPERIENCE

Research Intern

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist

CSIR-CEERI

Jan '24 – May '26

- Monocular vision-based UAV navigation for orchard monitoring using YOLOv11. [🔗](#)
- Multi-critic decomposition for off-policy reinforcement learning (TD3, SAC, DDPG) for autonomous indoor drone navigation. [🔗](#)

Research Intern

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor

MARMoT Lab, NUS

Mar '25 – Dec '25

- Uncertainty-guided path planning via conditional generative layout prediction (CogniPlan). [🔗](#)
- Vision-attention-driven autonomous navigation with semantic understanding, extending CogniPlan and ViNT.

Research Assistant

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor

BITS Pilani

Apr '24 – Oct '25

- Physics-Based Launcher Mechanical Design and FEM Analysis. [🔗](#)
- Computer Vision Pipeline for 3D Trajectory Reconstruction and Shot Classification.

Research Intern

Supervisor: [Dr. Pradeep Kundu](#), Assistant Professor

KU Leuven
Sep '24 – May '26

- Synthetic data generation for bearing fault diagnosis under data scarcity. [🔗](#)

Research Intern

Supervisor: [Dr. Rishi Verma](#), Head (ESDS), PPEMD

BARC
Jun '24 – Jul '24

- High-velocity impact testing and analysis of layered metal-composite shields using an electromagnetic railgun.
- Development and testing of a constrained detection and tracking algorithm for drones.

INDUSTRY EXPERIENCE

ML Intern

Supervisors: [Mr. Siddarth Malreddy](#), Tech Lead Manager & [Mr. Ishan Nigam](#), Senior ML Engineer

Uber
Jul '25 – Dec '25

- Augmented uLabel with object tracking and XGBoost-based anomaly detection for human-in-the-loop video annotation. [🔗](#)

TEACHING EXPERIENCE

Teaching Assistant

CS F320: Foundations of Data Science, BITS Pilani

Jan '26 – May '26

ME F218: Advanced Mechanics of Solids, BITS Pilani

Jan '25 – May '25

ME F216: Materials Science and Engineering, BITS Pilani

Sep '24 – Dec '24

LEADERSHIP & VOLUNTEERING EXPERIENCE

Deputy Director of Academic Programming

Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania

Aug '26 – Present

Vice-President

South Asian Association (Rangoli), University of Pennsylvania

Aug '26 – Present

President & Secretary

Mechanical Engineering Association (MEA), BITS Pilani

Jun '24 – May '26

President & Tech Fest Coordinator

Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani

Oct '24 – Jul '25

Project Manager

Tinkerer's Lab (TL), BITS Pilani

May '24 – Jul '25

Treasurer

Bengali Cultural Association (Moruchhaya), BITS Pilani

May '24 – Jul '25

AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) [🔗](#)

Apr '26

3rd Place — MathWorks Global Drone Student Challenge 2025

Mar '25

Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) [🔗](#)

Jul '24

Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 [🔗](#)

Jul '24

5th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)

Sep '23

PROFESSIONAL ACTIVITIES

Graduate Student Member — IEEE

2026

TECHNICAL SKILLS

Relevant Courses: Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins

Programming Languages: Python, C++, C, Shell (Linux)

Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Open3D, Matplotlib, Weights & Biases (W&B)

Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR

CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360